

House Price Prediction - Project Summary

Internship Project - Week 1 | Pranay Agarwal

Problem Statement

Real estate buyers and sellers often rely on guesswork or outdated comparisons to estimate a property's fair value. The goal of this project is to build a regression model that predicts house prices based on property features such as area, number of rooms, and amenities, and to identify which features most strongly influence price.

Dataset

Source: Kaggle - Housing Prices Dataset by YasserH

545 rows x 13 columns | Features: area, bedrooms, bathrooms, stories, mainroad,

guestroom, basement, hotwaterheating, airconditioning, parking, prefarea, furnishingstatus

Target variable: price

Data Cleaning Steps

- Checked for missing values - none found in the dataset
- Checked for duplicate rows - removed if any were found
- Converted binary yes/no columns to numeric (1/0)
- Applied one-hot encoding to furnishingstatus (multi-category column)
- Verified all features are meaningful for prediction via correlation analysis

Models & Results

Metric	Linear Regression	Random Forest	Winner
MAE	402,470	637,092	Linear Regression
RMSE	517,678	808,095	Linear Regression
R2 Score	0.9503 (95%)	0.8789 (88%)	Linear Regression

Data split: 80% training / 20% testing | Random state: 42

Key Findings

- Most influential features: area (0.520), bathrooms (0.231), stories (0.097)
- Area alone accounts for over 50% of the prediction power
- Bedrooms have surprisingly low importance compared to bathrooms and area
- Air conditioning and preferred area location add significant price premiums
- Linear Regression achieved the best R2 score of 0.9503 (95% accuracy)

Visualizations Created

- Chart 1: Histogram - Distribution of house prices with mean/median markers
- Chart 2: Correlation Heatmap - Feature relationships with price
- Chart 3: Scatter Plot - Actual vs Predicted prices for both models

Tools Used: Python 3.x | Pandas | Scikit-learn | Matplotlib | Seaborn | Jupyter Notebook

Business Recommendation

For real estate businesses, the data strongly suggests that maximizing usable area and adding bathrooms offers the highest return on investment. Properties with air conditioning and those in preferred areas fetch premium prices. Agents should highlight these features in listings, and developers should prioritize spacious layouts with modern amenities.